A novel multi-resolution method using wavelet packet transform and k-means clustering for analyzing data of passive cavitation detectors

Max Au-Yeung[1, 2] and Reza Haqshenas[1, 2]
[1)] *Ultrasonics Group, Department of Mechanical Engineering, University College London, London, UK*
[2)] *Ultrasonics Laboratory, Department of Mechanical Engineering, University College London, London, UK*[a)]

Acoustic emissions from cavitating bubbles contain complex information in the time and frequency domains. Passive cavitation detectors are used to measure these emissions. By processing this data, it becomes possible to extract cavitation features including broadband emissions, harmonics, sub- and ultra-harmonics. This allows monitoring cavitation which has diverse industrial applications. This paper presents a novel approach that utilizes the wavelet packet transform, statistical metrics and k-means clustering to detect and classify these features in acoustic emissions. These findings enable us to quantitatively separate the stable and transient features of acoustic cavitation. This method was validated on acoustic emissions obtained by solving a single bubble dynamics (Gilmore-Akulichev model) and subsequently tested on experimental data from biomedical ultrasound applications.

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I. INTRODUCTION

Acoustic cavitation refers to the formation, growth and oscillation of bubbles (cavities filled with gas and vapour) in a liquid using acoustic waves.[1] These bubbles may periodically oscillate whilst exposed to acoustic waves (known as stable cavitation) or grow and implode violently within a few cycles of excitation (known as inertial/transient cavitation). There are several physical effects associated with acoustic cavitation, including creation of free radicals, shock waves, shear stresses, microjets and micro-streaming in the insonated liquid.[1] To utilize these effects for different applications, ultrasound waves are widely used to produce acoustic cavitation, e.g. imaging ultrasound,[2,3] therapeutic ultrasound,[4–6] sono-chemistry,[7,8] and sono-crystallization.[9–11] To monitor these processes and to perform mechanistic studies, acoustic cavitation must be measured and evaluated quantitatively.

Passive[5,12] and active[13,14] cavitation detections are the two main techniques used in biomedical ultrasound to quantitatively measure acoustic emissions generated by bubbles. Passive techniques use a single element ultrasound transducer, i.e. passive cavitation detector (PCD), to acquire acoustic emissions whereas active techniques use more sophisticated ultrasound array transducers. Passive methods are cost-effective and easy to implement, hence they have been widely used in basic and applied research. Some examples in biomedical ultrasound are as follows: thermal ablation,[9] histotripsy,[15] drug and gene delivery,[16] blood-brain-barrier opening.[17,18] Several methods have been developed to analyze acoustic emissions from cavitations, including spectral techniques, combination of spectral and digital filters and/or singular-value decomposition, time-frequency methods including short time-Fourier transform, Bilinear time-frequency distributions, and wavelet transforms.[5,19–23] These works detect and classify the two main types of acoustic cavitations qualitatively. To carry out fundamental studies and elucidate the concomitant biological/physical effects produced by acoustic cavitation, advanced signal processing techniques are required to detect, extract, classify and quantitatively characterize the features of acoustic cavitation. To tackle this problem, we developed a method based discrete wavelet decomposition and spectral and statistical analyses to identify different types of cavitation and estimate their energies.[9] Due to limitations of discrete wavelet transform, this method cannot isolate cavitation features in complex scenarios where we have overlapping spectral signatures. In this paper, we present a new method which uses combined discrete wavelet packet transform, statistical signal processing and k-means clustering for quantitatively analyzing acoustic emissions, extracting and classifying cavitation features. The method is implemented in a Python library which is available in a GitHub repository.

[a)]; zcemauy@ucl.ac.uk

The method is discussed in detail in Section II. For testing and verifying the accuracy and efficiency of the method, synthetic acoustic emissions were generated by solving the problem of single bubble dynamics using Gilmore-Akulichev model, presented in Section II G. The method was then applied to analyze the theoretical acoustic emissions. Finally, the method was applied to process experimental PCD data, explained in Section III C 1.

II. MATERIALS AND METHODS

A. Features of acoustic cavitation

The signal from a PCD contains multiple features: (i) a range of harmonics with frequency of nf_0 where f_0 is the fundamental frequency and $n = 1, 2, 3, \ldots$ is an integer. At low powers, the fundamental frequency is the only component[24]. As the ultrasound amplitude is increased, but before collapse cavitation occurs, high-order harmonics are emitted by the bubbles. The emission with frequency equal to twice fundamental was prominent and had an amplitude proportional to the square of the fundamental[24]. (ii) sub-harmonics with frequency of f_0/n. These components can be formed by both stable and inertial cavitation[1]. The main hypothesis about the formation of the sub-harmonics is that they are generated by the non-linear pulsations of stable cavitation. If they are emitted by a system with only inertial cavitation, this can be explained by the prolonged expansion phase that can occur when the bubble oscillates for a few cycles before collapsing. The 1/2 and 1/3 frequency sub-harmonics are the most commonly occurring ones in acoustic cavitation. (iii)ultra-harmonics with frequency of mf_0/n where $m \neq n$ is an integer. These components can be formed by both stable and inertial cavitation[1]. (iv) broadband frequency component which is usually considered as an indicator of transient cavitation. The shock wave produced by an inertial collapse manifests as broadband noise in the spectrum[20]. In the case of a cloud of bubbles, the interaction between bubbles can produce harmonics and sub- and ultra-harmonics as well[25], further increasing the difficulty in analyzing the PCD data.

These features have different types of time-frequency structures. The (sub- and ultra-) harmonic components of the signal exist for a long time interval but they have localised frequency values equal to mf_0/n for integer m and n values. The transient components, however, have an in impulsive structure in time and relatively broadband structure in the frequency domain. The noise component (mainly measurement noise) exists over a long period and manifests as a broadband distribution in the frequency domain.

B. Wavelet packet transform

Wavelet packet bases were developed to effectively decompose a piecewise smooth signal with a varying time-frequency structure. Orthonormal wavelet packet bases divide the frequency axis into separate intervals of various sizes, known as scales.

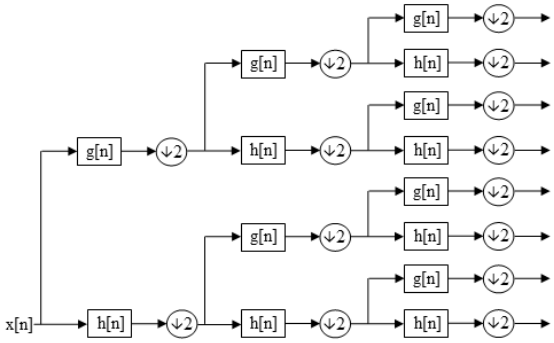


FIG. 1. The DWPT implemented as a filter bank

In fast orthogonal discrete wavelet transform (DWT), a bank of conjugate mirror filters and downsampling (by a factor of 2) are used to decompose a signal $x[n]$ of length $N = 2^{n_0}$ into a cascade of coarser approximations $a_j[n]$ and finer details $d_j[n]$ at a scale j[26]

$$\begin{aligned} a_{j+1}[n] &= \sum_{m=0}^{M-1} h[m]a_j[m-2n] \\ d_{j+1}[n] &= \sum_{m=0}^{M-1} g[m]a_j[m-2n] \end{aligned} \tag{1}$$

where $a_0[n] = x[n]$, $h[m]$ and $g[m]$ are the impulse responses of the filters corresponding to the scaling and wavelet bases functions and are related as follows

$$g[m] = (-1)^{1-m}h[1-m] \tag{2}$$

$g[m]$ and $h[m]$ can be viewed as transfer functions of low- and high-pass filters, respectively, with length M. Thus, approximations and details represent the amplitudes of low- and high- frequency components, respectively, of the decomposed signal. In Eq. (1) , the approximations (a_j) are only divided to low- and high-frequency components. Instead, we can iteratively split both approximation (a_j) and detail (d_j) at each scale, usually represented in a binary tree. This increases the computational complexity of discrete wavelet packet transform (DWPT) to $O(Nlog_2(N))$ compared to $O(N)$ for DWT. Any node of the binary tree includes a packet of wavelet transformed component, i.e., the coefficients of filtered signal at a frequency subband, and is labeled by tuple (j, k) where j is the scale number (the depth of the node in the tree) and k is the number of nodes that are on its left at the same scale (position of a node at a given depth). A schematic diagram of discrete wavelet packet transform using a bank of filters is shown in Fig. 1 .

The properties of the high- and low-pass filters are determined by the type and order of the wavelet function.

There is a trade-off between the time and frequency resolution when selecting the order of the wavelet. A higher-order wavelet leads to subbands with less overlap but may struggle to pick up closely-spaced features. Here, the effectiveness of wavelets is assessed using the silhouette score in the clustering algorithm.

Compared with discrete wavelet transform, DWPT provides higher frequency resolution (and decreased time resolution) at higher frequencies. The more levels of transform, the narrower the frequency subbands. We will use Shannon entropy minimization algorithm[27] to find the maximum scales of decomposition for each signal. In this algorithm, if the combined entropy of the child nodes is greater than that of the parent node, the nodes are deemed to have better information and are kept in the final wavelet packet basis. Using this measure of entropy, the algorithm will tend to split strong harmonics further while containing transient features at higher-level nodes. This reduces the redundancy in wavelet packet transform, providing a more compact representation of the signal. It also addresses the class imbalance that comes from the excessive number of nodes at the higher frequencies in the full WPT. This makes use of the primary strength of the WPT in the flexibility it provides in tiling the time-frequency domain.

The packets can be separately reconstructed into signals which combine to form the original signal. The stable and transient components of the signal tend to have different time-frequency signatures, so the transform will help separate them out. Once separated, the packets can be classified to give an indication of the presence and extent of each type of cavitation.

C. Statistical metrics for feature detection

To extract the signatures of inertial and stable cavitations at each subband, we will characterise the DWPT coefficients of nodes at the final (deepest) scale by the following statistical metrics: energy, kurtosis, crest factor, skewness, entropy and the logarithms of energy. The energy of a packet is a measure of how much the signal is concentrated within that subband, calculated as follows

$$E_{j,k} = \sum_{n=1}^{N_{j,k}} x_{j,k}[n]^2 \tag{3}$$

where $x_{j,k}[n]$ are coefficients in a subband (j,k) and $N_{j,k}$ is the number of coefficients in the subband. Note that using orthogonal wavelet bases for DWPT preserves the energy of the signal. Hence, the sum of the energy of all nodes in the final scale equals to the energy of the time series signal. The average energy of the coefficients in a node is defined as

$$\bar{E}_{k,j} = \frac{1}{N_{k,j}} E_{j,k} \tag{4}$$

where $E_{j,k}$ is the energy of the coefficients in a subband (j,k) and N_k^j is the number of coefficients in the subband.

To better distinguish fundamental, sub- and ultra-harmonics, we also use the log-energy metric, i.e., $LE_{j,k} = \log E_{j,k}$. To measure the energy distribution of a subband - compared to other subbands, we use Shannon entropy which is defined as

$$S_{j,k} = -\sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{j,k}} \log \frac{x_{j,k}[n]^2}{E_{j,k}} \tag{5}$$

where $E_{0,0}$ is the total energy of the signal.

Level entropy is a variation on the calculation of entropy given by

$$S'_{j,k} = -\sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{0,0}} \log \frac{x_{j,k}[n]^2}{E_{0,0}} \tag{6}$$

We will use level entropy to also obtain the minimum entropy decomposition.

The spectral entropy is a measure of the complexity of the distribution in the FFT of the subband. This calculated using Eq. (5) where $x_{j,k}[n]$ is replaced with its magnitude spectrum and $E_{j,k}$ is the energy of the magnitude spectrum. The lower the spectral entropy of a subband, the more deterministic components (information about harmonics) are within the subband. The transient components have a broadband frequency signature, hence a higher entropy can be attributed to transient and/or noise components of the signal. To distinguish these features we need additional statistical metrics.

Kurtosis is a measure of how often outliers occur in a signal. It is viewed as a metric for detecting transient cavitation. Kurtosis is calculated using the following relationship

$$K_{j,k} = \frac{1}{n\sigma_{j,k}^4} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \overline{x}_{j,k})^4 \tag{7}$$

where $\overline{x}_{j,k}$ and $\sigma_{j,k}$ are the mean and standard deviation of the coefficients of the node (j,k).

Skewness measures the extent to which a distribution is asymmetrical. It is given by

$$Sk_{j,k} = \frac{1}{n\sigma_{j,k}^3} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \overline{x}_{j,k})^3 \tag{8}$$

The absolute value of the skewness of the coefficients is expected to be larger in magnitude when the signal has transient components.

The crest factor is the ratio between the peak value and the mean of the signal, indicating how extreme the peaks are in the signal. It is obtained by

$$C_{j,k} = \frac{\max\left(|x_{j,k}[n]|\right)^2}{x_{rms}^2} \tag{9}$$

where x_{rms} is the root-mean-square of $x_{j,k}[n]$.

A feature matrix of these statistical metrics of all subbands is formed. The next step of the algorithm is to classify these features, hence subbands. A commonly used technique, adopted in our algorithm too, is k-means clustering[28,29], which is discussed in the next section.

D. k-means clustering for feature classification

k-means splits a dataset comprised of n observations, each with a number of features, into k clusters. The aim of the algorithm is to minimise the total loss measured as the sum of the Euclidean distances between the centroid of the cluster and each point assigned to it (see Eq. (10)). To do this, the cluster centroids are initiated randomly and each data point is assigned to the nearest cluster. The clusters are iteratively re-allocated to minimise the loss function.

$$\Theta = \sum_{i=1}^{n} \sum_{j=1}^{k} \rho^{(i,j)} \|x^{(i)} - \mu^{(j)}\|^2 \tag{10}$$

where Θ is the distortion (loss) function, ρ is the assignment of the data point i to cluster j, and x and μ are the data point and cluster centroid respectively.

k-means is a non-convex optimisation problem and yields different results depending on the starting position of the clusters. This is further discussed in section II E.

Another issue of the k-means algorithm is that the number of clusters k must be specified before training. This presents a challenge in this scenario where the number of clusters is expected to vary depending on the type of cavitation that occurs. For example, a signal comprised of a sine wave and noise, representing a measurement of stable cavitation, will results in nodes which fall into two clusters: those representing noise, and those representing the fundamental frequency. Once harmonics and transients are added to the signal, more clusters could be formed. Therefore, the algorithm must adapt to the number of clusters that exist in the data. Several techniques have been developed for choosing the optimal number of clusters[30,31]. Here, we will use the silhouette score to to automate the selection of k. It is easy to implement and works well for small datasets.[32]. The silhouette score is score is given by

$$s = \frac{b - a}{\max(a, b)} \tag{11}$$

where b is the mean distance between a point and the points in the nearest cluster a is the mean distance between the point and all other points in the same cluster. The nearest cluster is found by finding the cluster centre that is closest the the point. The metric used to select k is the average silhouette score for every point in the dataset. We will iteratively change k to maximise the silhouette score.

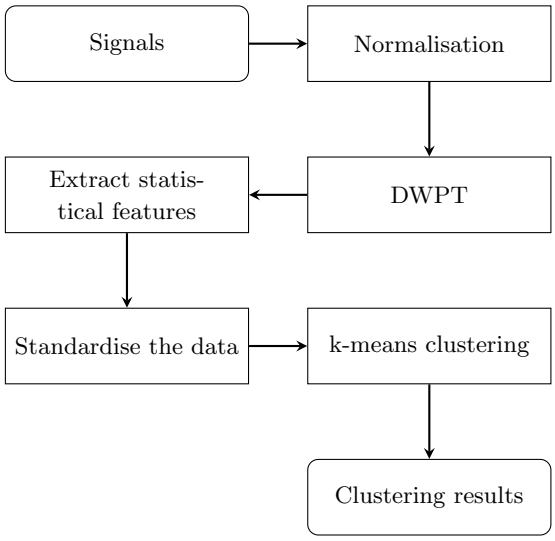


FIG. 2. Flowchart of proposed method

E. Method overview

The proposed algorithm is outlined in Fig. 2 . The PCD data were normalised by taking off the mean and dividing by the standard deviation. Next, the DWPT was applied to the signal, decomposing it into frequency subbands using the minimum entropy decomposition algorithm and the statistical metrics were calculated at the leaf nodes. The nodes with frequency higher than a specified frequency were left out. These metrics were normalised using a standard scalar, fitting each set of statistical metrics to a standard Guassian distribution to ensure that their difference in scale does not affect clustering. Each node, now represented as a set of features calculated from their coefficients, was fed into the k-means algorithm for to be clustered. We only need to specify the wavelet family and the order of the wavelet function. The depth of decomposition and the number of clusters was determined automatically.

F. k-means parameters

The k-means algorithm was run with Lloyd's algorithm described in section II D. The clustering problem is sparse, so the effects the non-convexity of the k-means algorithm was mitigated using multiple runs of the k-means algorithm. The clustering with the best value of the distortion function (see Eq. (10)) was selected. Furthermore, the k-means++ algorithm[33] is used to select the starting centroids. This method selects data points as clusters one-by-one, while enforcing that points further away from existing selections are more likely to be chosen. This can improve clustering time and performance, reducing the chance that clusters are merged.

G. Gilmore–Akulichev model

Gilmore's model describes the dynamics of a spherical bubble in a compressible liquid, undergoing symmetric oscillations due to acoustic excitations. The model has been validated against experimental data and widely used for studying acoustic cavitation.[34] The Gilmore equation is expressed as follows[35]

$$R\ddot{R}\left(1 - \frac{\dot{R}}{C}\right) + \frac{3}{2}\dot{R}^2\left(1 - \frac{\dot{R}}{3C}\right) = H\left(1 + \frac{\dot{R}}{C}\right) + \frac{R\dot{H}}{C}\left(1 - \frac{\dot{R}}{C}\right) \quad (12)$$

where R is the instantaneous bubble radius, C and H are the speed of sound in the liquid at the bubble wall, and the enthalpy difference in the liquid at the bubble wall (at pressure $P(R)$) and at infinity (at P_∞), respectively. Time derivatives are designated by dots. Pressure at infinity is the sum of the static reference pressure P_0 and the acoustic pressure $P_{ac}(t)$, i.e., $P_\infty(t) = P_0 + P_{ac}(t)$. Gilmore's model implicitly assumes that the acoustic pressure is applied instantaneously and uniformly around the bubble without specifying a source location in the liquid. Thus, the liquid density and speed of sound at infinity are taken at the static reference state, denoted by ρ_0 and c_0, respectively, below. C and H are calculated using the following equations

$$C = \sqrt{c_0^2 + (n-1)H} \quad (13)$$

$$H = \frac{nA^{1/n}}{(n-1)\rho_0}\left[(P(R) + B)^{(1-1/n)} - (P_\infty(t) + B)^{(1-1/n)}\right] \quad (14)$$

where $A, B,$ and n are constants in the Tait's equation of state which are obtained experimentally. The time derivative of the enthalpy in the liquid at the bubble wall becomes

$$\dot{H} = \frac{A^{1/n}}{\rho_0}\left[\dot{P}(R)\,(P(R) + B)^{-1/n} - \dot{P}_{ac}(t)\,(P_\infty(t) + B)^{-1/n}\right]. \quad (15)$$

Note that $\dot{P}_\infty(t) = \dot{P}_{ac}(t)$. $P(R)$ is calculated using a modified Laplace equation

$$P(R) = \left(P_0 + \frac{2\sigma}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} \quad (16)$$

where R_0 is the initial bubble radius, γ, μ and σ are the adiabatic index (ratio of specific heats), dynamic viscosity, and surface tension, respectively. Finally, taking the time derivative of the above equation, we arrive at

$$\dot{P}(R) = -3\gamma\frac{\dot{R}}{R}\left(P_0 + \frac{2\sigma}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\gamma} + \frac{\dot{R}}{R}\frac{2\sigma}{R} + 4\mu\left(\frac{\dot{R}}{R}\right)^2 - \frac{4\mu\ddot{R}}{R}. \quad (17)$$

Eqs. 13-17 are used to numerically solve the Gilmore equation Eq. (12) . The radiated pressure in the liquid is then given by the Akulichev formula

$$P_r(t) = A\left(\frac{2}{n+1} + \frac{n-1}{n+1}\left(1 + \frac{n+1}{rc_0^2}G\right)^{1/2}\right)^{2n/(n-1)} - B \quad (18)$$

where r is the distance from the centre of the bubble, and $G = R(H + 0.5\dot{R}^2)$. This equation assumes that acoustic emissions from an oscillating bubble propagate as outward traveling spherical waves. The Gilmore–Akulichev model was nondimensionalized and solved numerically using a stiff ODE solver for different sonication conditions. The implementation is included in the project's GitHub repository.[36] The implementation and numerical results were verified against published results for identical parameters.

III. RESULTS AND DISCUSSION

This section presents the results of studies aiming to test the algorithm's ability for identifying and classifying the features of stable and transient cavitation. Initially, the algorithm will be applied to simulated data generated by solving the Gilmore–Akulichev model for parameters resulting in stable and transient cavitation. Subsequently, we will process PCD data obtained experimentally, presented in Section III C.

A. Bubble dynamics data

The temporal description of the acoustic pressure is modelled as a linear sine wave, i.e., $P_{ac}(t) = P_A sin(2\pi ft)$ where P_A and f are the amplitude and frequency of acoustic pressure field. The frequency and pulse length are set to 2 MHz and $10/f$, to mimic experiments, and $P_A = 1$ MPa for the results presented below. We choose water and oxygen for the liquid and the gas. The physical parameters used for numerical simulations are as follows (at room temperature 293 K and atmospheric pressure $p_0 = 100$ kPa): $\rho_0 = 10^3\,\mathrm{kg\,m^{-3}}$, $\sigma = 7.25 \times 10^{-2}\,\mathrm{N\,m^{-1}}$, $\mu = 10^{-3}\,\mathrm{Pa\,s}$, $c_0 = 1482\,\mathrm{m\,s^{-1}}$, $n = 7$, $A = 3.04 \times 10^2$ MPa, $B = 3.039 \times 10^2$ MPa and $\gamma = 1.4$. The initial conditions are: $R(0) = R_0$ and $\dot{R}(0) = 0$. Here, we consider two scenarios in which $R_0 = 1\,\mu\mathrm{m}$ and $R_0 = 10\,\mu\mathrm{m}$, resulting in stable and transient cavitation, respectively.

The $R(t)$ curves and radiated pressures calculated at $r = 100\,\mu\mathrm{m}$ are shown in Figs XX.

B. Feature Selection

The features space is explored using self-organising maps[37], a type of neural network which can be used to visualise high dimensional data in two dimensions. The algorithm generates nodes which are arranged in a two-dimensional space, each containing an array of weights, one for each input feature. The nodes can be thought of as cluster centres which are interconnected such that neighbouring neurons have similar features. The features can be plotted separately, giving a set of maps which can be visually compared to evaluate the usefulness of each feature. The details of the algorithm will not be discussed here and can be found in paper written by Haroun et al.[38] in the context of bearing fault detection and diagnosis.

The data used to train the self-organising map uses 192 signals generated at a sampling rate of 50 MHz, representing both stable and transient cavitation. The signals have noise added to them and are windowed and normalised as described in Fig. 2 . The results are shown in 3. When the maps are visually similar, their respective features are correlated with each other. This shows that the high energy DWPT nodes tend to have low skewness, kurtosis, and spectral entropy, as expected given the physical interpretations of each. This area represents packet nodes containing fundamental frequencies. The spectral entropy and level entropy are able to find nodes around the high energy area. These represent harmonics which are not easily picked out using average energy alone. The nodes with high frequency and kurtosis, and low energy indicate signals with sharp transients. The rest of the high-frequency nodes represent noise.

Several plots look similar to each other, indicating redundancy in the feature space. Skewness, kurtosis, and crest factor perform similarly, while level entropy (normalised by the energy of the original signal), and spectral entropy are highly correlated.

Overall, the three best features are hand picked as average energy, spectral entropy, and kurtosis as they are useful for separating nodes representing fundamentals, harmonics, and noise.

This method was chosen over PCA which requires the computation of every feature at run time. It is worth noting that the results for self-organising maps is highly inconsistent where the shape of the maps can significantly change depending on the randomly-selected starting points. However, the observed correlations between the features remain similar each time.

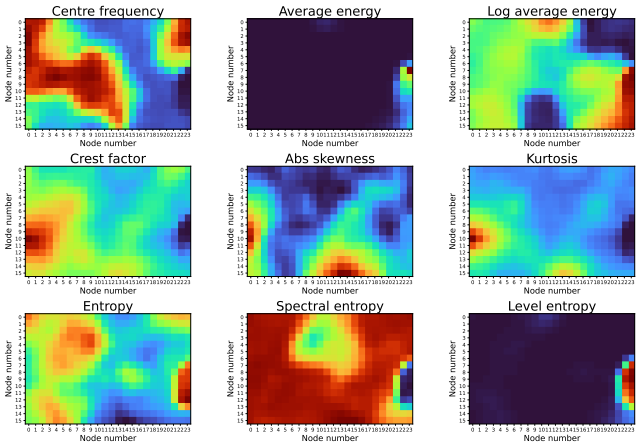


FIG. 3. Self organising maps results where the nodes are arranged in a rectangle of aspect ratio 1.5. Each feature has its own plot to show its relationship with other features. The data was generated using a minimum entropy decomposition of data generated using the Gilmore-Akulichev model. The minimum and maximum levels of decomposition are 5 and 10 respectively. The db10 wavelet is used.

C. Experimental acoustic emissions

1. Experimental set-up and protocol

2. Analyzing PCD data

IV. SUMMARY

In this paper a novel approach was developed for classifying PCD data by clustering DWPT nodes, in an attempt to create an automated algorithm that can separate out features relating to stable and transient cavitation. The DWPT has been found to be useful in separating out these components in a flexible manner, especially when combined with the minimum entropy decomposition, which is able to generate a reasonable time-frequency tiling. Several features were assessed to help classify the nodes. Self-organising maps were used to help visualise the feature space, a useful method in determining which ones may be the most useful. However, the algorithm is unable to consistently separate harmonics and sub-harmonics from lower-kurtosis noise. This leads to several suggestions for future work:

- Explore alternative algorithms to k-means which may be able to find more meaningful clusters.
- Explore other methods of generating training data with more involved models.
- Expand the work to cavitation in other common mediums like tissue.
- Separate noise generated through cavitation and measurement noise.
- Extend the algorithm to automatically detect cavitation activity over noise. This can be useful for applications where the ultrasound is pulsed.
- Explore the use of entropy, getting around the issue discussed in section III B where the level of the transform dominates their measure. Possible approaches could involve calculating the entropy and spectral entropy on the reconstructed signals which always have the same length as the original signal or by using an undecimated transform.

- The sparseness of the algorithm could be addressed by clustering nodes from more than one sample. This could allow it to classify harmonics in separate clusters.

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